Is AI Technical Change Different for the Equilibrium Real Interest Rate?
A Flex-Price Ramsey Summary of the First Decade

Research note (draft)

October 2026

Abstract

In a flex-price Ramsey economy, the real interest rate over the first decade of an AI-driven technology shift is set by one thing: how fast consumption is expected to grow relative to trend. Three forces change that. *Embodiment* decides whether consumption gains come early or late in the decade; it is the largest force. *AI versus regular capital* matters mainly through how much investment AI capital absorbs, and much less through the labor share: even with half of households hand-to-mouth, the labor-share fall moves the decade-average rate by at most 0.1 percentage points. The *elasticity of saving* scales every response without changing the ranking: an intertemporal elasticity of 0.5 or 0.2 multiplies the first-decade effect by about 1.7 or 3. Habits matter mainly after surprise windfalls. The *level* of adjustment costs matters more than how they are matched across capital stocks. All shocks are one-time level shifts that raise long-run output by 10 percent, so the long-run rate is unchanged in this model by construction. Whether AI changes it depends on saving behavior, which this note holds within Ramsey.

1 The one-line intuition

With a representative Ricardian household and CRRA preferences ($\theta = 1/\text{IES}$), the Euler equation gives

$$\ln(1 + r_{t+1}) - \ln(1 + r_0) = \theta\, \Delta \ln \tilde{c}_{t+1}, \qquad \tilde{c} = \frac{c}{Z},$$
$$\sum_{t=0}^{9} \big[\ln(1 + r_{t+1}) - \ln(1 + r_0)\big] = \theta \big[\ln \tilde{c}_{10} - \ln \tilde{c}_0\big]. \tag{1}$$

The real rate in the first decade is θ times the expected *catch-up* of consumption against trend. (The identity is exact without habits; the numerical solutions satisfy it to 10^{-14}.) The rate is high when a consumption gain is coming but has not yet arrived. It is low when the gain has already arrived, or when it is far off. So the questions become: how much of the consumption gain falls inside the decade, and how much does θ amplify it?

What is held fixed.

- Cobb-Douglas capital-labor bundle, AI capital as a substitute ($\varepsilon = 1.5$), fixed hours, trend growth 1.5%, initial $r = 7.5\%$.

- Equal depreciation for both capital stocks, $d_A = d_K = 6.6\%$.
- Adjustment costs $P_j \frac{c_j}{2}(i_j - x_j)^2 K_j$, with $c = 1/x$ for regular capital (a unit elasticity of investment to Q).
- Every shock is a permanent level shift in technology, sized for a *10% long-run output gain*. Regular-capital progress needs a 29% improvement and AI progress a 3.4-fold improvement.
- Two timings. An *S-curve*, $\frac{1}{2}[1 - \cos(\pi t/10)]$, is announced at $t = 0$ and complete by year 10. A *jump* is a surprise at $t = 0$.
- The summary statistic is the *decade-average rate*: the mean of $r_1, \ldots, r_{10}$ minus 7.5%, in percentage points.

2 Results

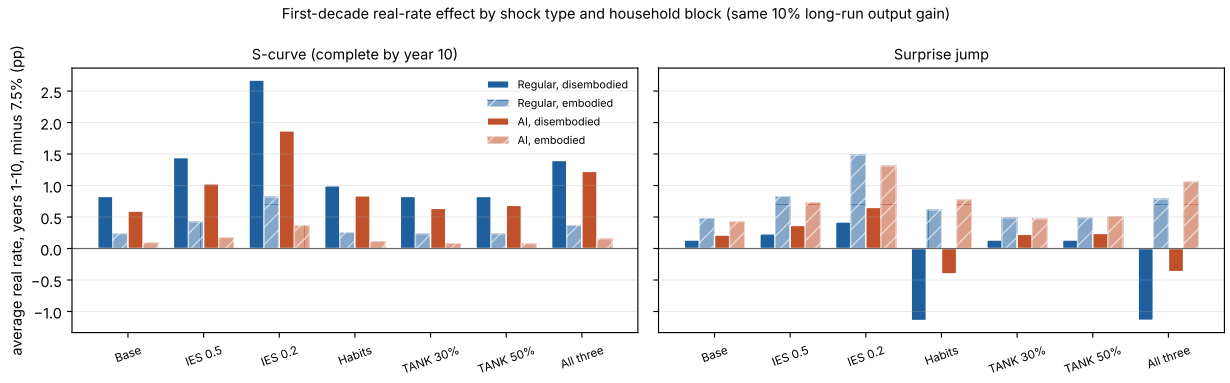


Figure 1: Decade-average real rate relative to 7.5%, by shock type (solid: disembodied; hatched: embodied; blue: regular capital; orange: AI) and household block.

Table 1: Decade-average real rate minus 7.5% (pp). "IES" columns change θ only. "Habits" is $h = 0.7$ with $\theta = 1$. "TANK" is the hand-to-mouth share. "All three" is IES 0.5, $h = 0.7$ and 30% hand-to-mouth.

		Base	IES 0.5	IES 0.2	Habits	TANK 30%	TANK 50%	All three
S-curve	Regular, disembodied	+0.83	+1.44	+2.67	+1.00	+0.83	+0.83	+1.40
	Regular, embodied	+0.24	+0.42	+0.82	+0.26	+0.23	+0.23	+0.37
	AI, disembodied	+0.59	+1.03	+1.87	+0.84	+0.64	+0.69	+1.22
	AI, embodied	+0.09	+0.18	+0.37	+0.12	+0.09	+0.08	+0.15
Jump	Regular, disembodied	+0.13	+0.23	+0.42	-1.14	+0.13	+0.13	-1.14
	Regular, embodied	+0.48	+0.83	+1.49	+0.61	+0.49	+0.49	+0.79
	AI, disembodied	+0.21	+0.37	+0.65	-0.40	+0.23	+0.24	-0.37
	AI, embodied	+0.43	+0.73	+1.31	+0.78	+0.47	+0.52	+1.06

2.1 Embodiment: when does the consumption gain arrive?

Take the base column of Table 1 and the first part of Table 2.

Table 2: Base household. $\Delta \ln c$ is the gain in detrended consumption from year 0 to 10 (percent); the next column is the right-hand side of (1) divided by 10. Wages and output are log deviations (percent) at year 10; labor share in percent.

		$\Delta \ln c$, yr 0→10	$\theta \Delta \ln c/10$	r_1	peak r (yr)	$\ln W$, yr 10	$\ln Y$, yr 10	labor share, yr 10
S-curve	Regular, disembodied	7.7	+0.77	7.69	8.75 (5)	7.7	7.6	69.7
	Regular, embodied	2.2	+0.22	7.51	7.99 (9)	2.2	1.9	69.7
	AI, disembodied	5.5	+0.55	7.58	8.41 (6)	4.0	5.8	68.3
	AI, embodied	0.9	+0.09	7.50	7.89 (9)	1.2	1.0	69.2
Jump	Regular, disembodied	1.3	+0.13	7.67	7.67 (0)	8.3	8.2	69.7
	Regular, embodied	4.5	+0.45	8.14	8.14 (0)	4.4	4.2	69.7
	AI, disembodied	2.0	+0.20	7.76	7.76 (0)	4.5	6.4	68.3
	AI, embodied	4.0	+0.40	8.06	8.06 (0)	2.5	3.3	68.8

- *Anticipated shift (S-curve), disembodied.* The technology improves along the S-curve and consumption follows it, rising 7.7% (regular) or 5.5% (AI) over the decade. The rate rises 0.8 and 0.6 points on average and peaks near year 5–6 at 8.8% and 8.4%.
- *Anticipated shift, embodied.* The gain requires building capital first. Investment is front-loaded and consumption is held down. Consumption gains only 2.2% (regular) or 0.9% (AI) by year 10, so the decade-average rate rises only 0.2 and 0.1 points. (Most of the catch-up falls after year 10.)
- *Surprise jump, disembodied.* Output and consumption jump at once, so little growth is left to expect (1.3% for regular capital): the average is +0.13 points.
- *Surprise jump, embodied.* Consumption starts low during the investment boom and recovers by 4.5%: +0.48 points, with a first-year rate of 8.1%.

The ranking reverses with timing. Disembodied progress raises r most when the shift is anticipated, and embodied progress raises it most when the shift is a surprise.

2.2 AI versus regular capital: how much comes from the labor share?

With the same 10% output gain, AI progress changes the first-decade rate less than regular-capital progress for an anticipated shift (0.59 vs 0.83 disembodied; 0.09 vs 0.24 embodied). It changes it a bit more after a disembodied jump (0.21 vs 0.13) and a bit less after an embodied jump (0.43 vs 0.48).

Why. AI progress delivers a smaller consumption gain per unit of output gain. Output has to cover the accumulation of a much larger AI capital stock (3.4-fold technology improvement; capital share rises from 30.4 to 32.2 percent). The wage gain is also smaller, 7.4% instead of 10% in the long run (Table 2), and the labor share falls by 1.8 points.

Does the labor share itself matter? It does not enter the representative-household Euler equation. To test it, the "TANK" columns make 30% or 50% of households hand-to-mouth, consuming their wages while Ricardians own all capital. A falling labor share then lowers hand-to-mouth consumption relative to output, so Ricardian consumption grows faster and r is higher. The effect is small. For the anticipated disembodied AI shift, the decade-average rate rises from 0.59 to 0.64 (30%) or 0.69 (50%). Regular capital, where the labor share is unchanged, is unaffected, and embodied AI barely moves. Within a decade, the labor-share fall is too small and too slow to be

a main channel (it is 1.8 points of income in total, and under embodied progress it comes mostly after year 10).

2.3 Elasticity of saving: how much does the IES amplify?

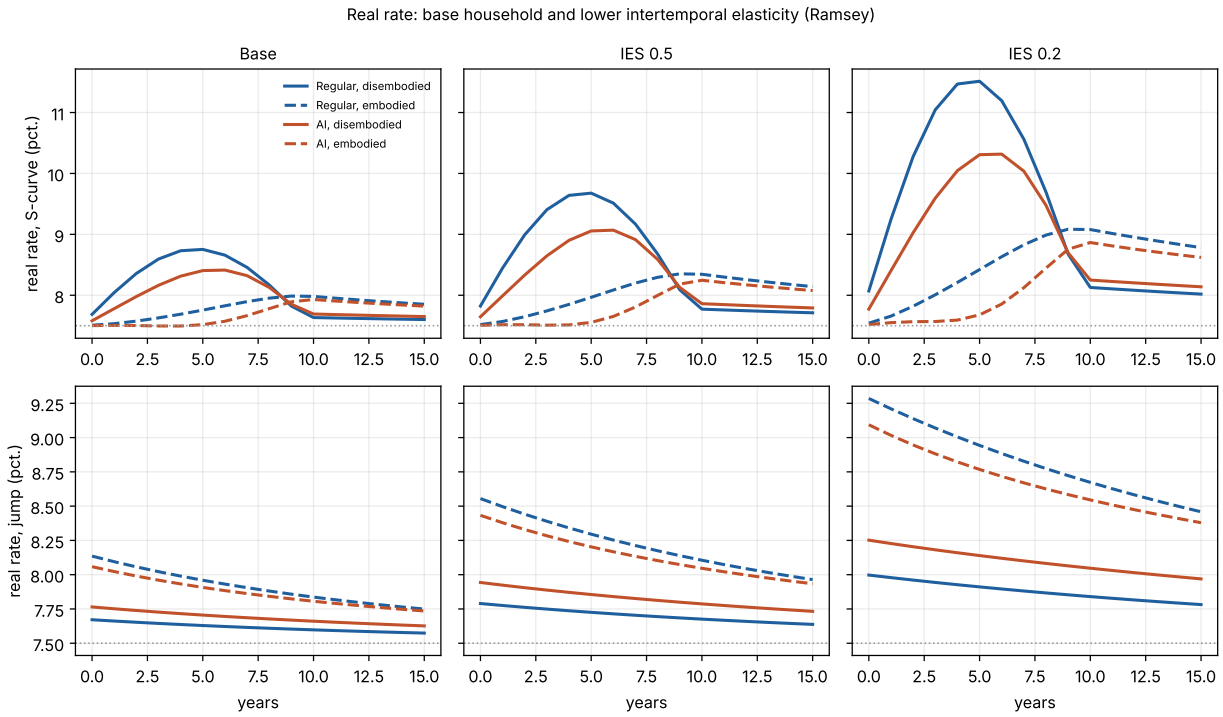


Figure 2: Real-rate paths with $\theta = 1, 2, 5$ (IES 1, 0.5, 0.2).

By (1), a lower IES multiplies the rate response to a given consumption path by θ. But households also tilt consumption less, so the consumption catch-up shrinks (for disembodied regular capital, from 7.7% to 6.7% and 4.9%). The net effect is roughly proportional to $\theta^{0.7}$: the decade-average rate rises by a factor of 1.7 at IES 0.5 and 3.2 at IES 0.2. For the disembodied regular S-curve, the peak rises from 8.75% to 9.7% and 11.5%. The ranking across embodiment, timing and AI versus regular does not change. A low IES also slows the whole adjustment: year-10 output gains are smaller (for embodied AI, 1.0% falls to 0.2%). So the standard log-utility baseline probably understates both the size and the persistence of real-rate movements.

2.4 Habits and hand-to-mouth households

- *Anticipated shifts* are only mildly affected by habits ($h = 0.7$ raises the decade average by 0.02–0.25 points), because consumption is already smooth. Their short-run effect is to make the response more hump-shaped.

- *Surprise disembodied jumps* are strongly affected. The windfall cannot be invested quickly (adjustment costs), households resist raising consumption (habits), so r must fall to deter saving. The first-year rate falls to 5.2% at $h = 0.3$ and 0.8% at $h = 0.7$. This is an artifact of a surprise windfall plus costly investment, so I would not rely on it quantitatively.

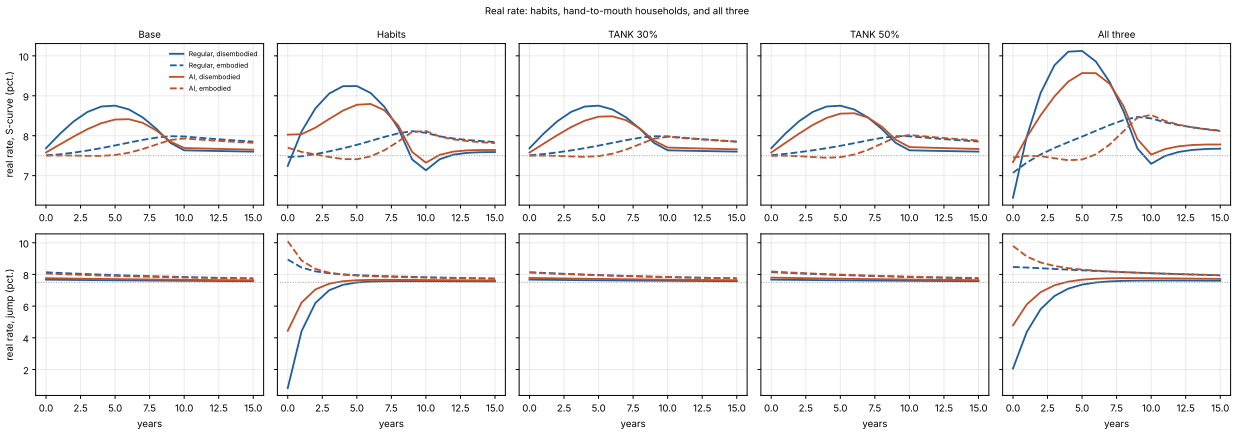


Figure 3: Real-rate paths with habits, hand-to-mouth households, and all three together.

- *Hand-to-mouth households* change very little, as discussed above.

2.5 Calibrating adjustment costs "similarly"

Equal depreciation and $c = 1/x$ give both capital stocks the same investment response to Q per unit of capital. The remaining question is whether the two adjust at the same *speed*. I calibrated AI's adjustment cost so that AI capital converges to a small shock at the same rate as regular capital. The two rates are already close with the same c (5.7% a year for AI versus 5.5% for regular capital), because the curvature of marginal products is similar. The matched c_A is 13.1 against $c_K = 12.3$, and the results match the same-c results to within 0.01 points (Table 3).

Table 3: Decade-average real rate minus 7.5% (pp) for alternative adjustment-cost calibrations, base household.

		equal speed	same c	c x 0.5	c x 2	none
S-curve	Regular, disembodied	+0.83	+0.83	+0.77	+0.86	+0.54
	Regular, embodied	+0.24	+0.24	+0.09	+0.33	-0.55
	AI, disembodied	+0.59	+0.60	+0.58	+0.60	+0.49
	AI, embodied	+0.09	+0.09	+0.05	+0.11	-0.14
Jump	Regular, disembodied	+0.13	+0.14	+0.20	+0.08	+0.45
	Regular, embodied	+0.48	+0.48	+0.69	+0.31	+1.48
	AI, disembodied	+0.21	+0.22	+0.31	+0.14	+0.72
	AI, embodied	+0.43	+0.44	+0.55	+0.32	+1.09

What matters is the *level* of c, not how it is matched. Without adjustment costs, embodied S-curve shifts *lower* the first-decade rate (-0.55 regular, -0.14 AI) because firms wait for cheaper capital. Embodied jumps raise it more ($+1.48$, $+1.09$). Adjustment costs smooth investment, which removes the wait-for-cheaper-capital response. With no adjustment costs, the AI-versus-regular ranking for an embodied S-curve flips (AI -0.14 above regular -0.55), because the dip is deeper for regular capital. With any positive c, AI is below regular capital.

3 Answering the motivating question

1. *Is AI technical change different for the equilibrium real interest rate?* In this flex-price Ramsey benchmark, the first-decade effect is smaller than for regular capital when the shift is anticipated, by about 0.15–0.25 points, because AI absorbs more of the output gain in accumulation and delivers less consumption growth. After a surprise jump, the sign depends on embodiment. The difference is modest next to the effect of embodiment, timing and the IES.

2. *Does it depend on the evolution of income shares?* Within the first decade, only weakly. The labor-share fall is about 1.8 points, mostly after year 10 for embodied progress, and it moves the decade-average rate by at most 0.1 points even with half the population hand-to-mouth. In the long run, income shares do not move r in Ramsey at all, because the Euler equation pins it. The earlier comparison with Solow, Diamond and annual-OLG saving shows that the income-share channel would matter for long-run r only if saving is less elastic than Ramsey implies.

3. *What would be needed to make shares matter?* Saving that depends on the type of income, and a labor-share fall that is large and early.

Caveats. Hours are fixed. All shocks are level shifts sized for the same long-run output gain, not shocks of equal technical size. The habit specification is additive and internal, with a strong annual h. The adjustment-cost level is unobserved, and Table 3 shows it is the most important calibration choice here for embodied progress. Nominal rigidities, the next step, would turn the response of r to consumption growth into a response of output as well.

A Verification

All results come from `ramsey_variants.py` (built on `capital_cases.py`). Every transition is a nonlinear perfect-foresight solve of the full equilibrium. The model uses the Euler equations of both capital stocks with the household block above, 250 years, with residuals below 10^{-10}. The pre-shock balanced growth path satisfies the Euler equations for every household block. For $h = 0$, (1) holds to 10^{-14}, and every shock reaches the 10 percent output gain to within 10^{-4} (the slowest case is IES 0.2).

Check	max abs. error
residual	7.7e-11
long-run ln Y = 10%	8.6e-05
identity sum ln(1+r) = theta dln c	1.3e-15
BGP satisfies the Euler equations	5.5e-14